

Find the general solution of each differential equation.

$$y) \frac{dy}{dx} = \frac{y}{x}$$

$$y) \frac{dy}{dx} = \frac{1}{\sec^2}$$

$$y) \frac{dy}{dx} = 3e^{x-y}$$

$$y) \frac{dy}{dx} = \frac{2}{e}$$

$$y) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = \frac{1}{2}$$

$$y) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln$$

$$y) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3)$$

$$y) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \sqrt[3]{e}$$

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$$\begin{aligned}
 & \tan y = x + \tan^{-1}(x + \sqrt{3}) \\
 & y = \ln(3e^x + 1) \\
 & C_1 y + \frac{1}{3} y = \ln(2e^x + 3) \\
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$$g) \frac{dy}{dx} = \frac{2x}{y^2}, y(2) = 3$$

$$h) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln 2$$

$$i) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = \frac{\pi}{4}$$

$$j) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \sqrt[3]{e}$$

$$k) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$l) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = 0$$

$$m) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

$$n) \frac{dy}{dx} = \frac{1+x^2}{y^2}, y(-1) = 1$$

$$\begin{aligned}
 & \frac{4}{3} \sqrt{2\sqrt{4}} \quad \text{and } y = x + \tan^{-1}(x + 1) \quad y = \ln(3e^x + 7) \wedge \\
 & + \frac{1}{3} y = \ln(2e^x + 3) \quad y = \tan^{-1}(x - 3)e^x \\
 & \cdot 3 y = \cos^{-1}(x - 1) \frac{e^{x-y}}{2} = \frac{x^3 - 2x + 1}{2}, \sqrt[3]{x} - \frac{1}{2} \\
 & \frac{1}{3} - x + \frac{1}{2} \sqrt{3} \quad \frac{1}{1 + e^{2y}} - \frac{1}{2\sqrt{3}} \quad 1
 \end{aligned}$$

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$$y = \ln(3e^x + 7) \wedge$$

$$\therefore \Delta^2 y = 2$$

$$\ln(2x^2)$$

$$+ \frac{1}{3}y' = \ln(2e^x + 3)$$

$$y = \tan^{-1}(\bar{x} - \bar{3})e^{\bar{x}}$$

$$y = \cos^{-1} \left(\frac{e^{2y} - 1}{e^{2y} + 1} \right) = x - \frac{\pi}{2}$$

$$\frac{1}{3} - \frac{1}{6}$$

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$$\begin{aligned}
 & \text{Let } y = x + \tan^{-1}(x + \sqrt{1+x^2}) \\
 & \text{Then } y = \ln(3e^x + 1) + \ln(2x^2 + 1) \\
 & \text{Also } y = \tan^{-1}(x + \sqrt{1+x^2}) \\
 & \text{Then } y = \cos^{-1}\left(\frac{e^{-x}}{2}\right) = \frac{1}{2} \ln \frac{1 + \sqrt{1+x^2}}{1 - \sqrt{1+x^2}} \\
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$$y = \ln(3e^x + 1) \wedge$$

$$\therefore \Delta^2 y = 2$$

$C_1 y$

$$= 2e^+ +$$

$$\ln(2x^2)$$

$$+ \frac{1}{3}y' = \ln(2e^x + 3)$$

$$y = \tan^{-1}(\bar{x} - 3)e^{\bar{x}}$$

$$y = \cos^{-1} \left(\frac{e^{2y} - 1}{e^{2y} + 1} \right) \equiv \frac{\pi}{2} - \frac{y}{2}$$

$$\frac{1}{3} - \frac{1}{7}$$

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$$2) \frac{dy}{dx} = \frac{1}{\sec^2}$$

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$$4) \frac{dy}{dx} = \frac{2}{e^y}$$

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$$6) \frac{dy}{dx} = 2e^{x-y}, y(-3) = \ln 2$$

$$7) \frac{dy}{dx} = \frac{1}{\sec^2 y}, y(3) = \frac{\pi}{4}$$

$$8) \frac{dy}{dx} = \frac{e^x}{y^2}, y(-1) = \sqrt[3]{e}$$

$$9) \frac{dy}{dx} = -\frac{1}{\sin y}, y(3) = \frac{\pi}{2}$$

$$10) \frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2) = 0$$

$$11) \frac{dy}{dx} = \frac{3x^2}{e^{2y}}, y(1) = 0$$

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$$3 y = \cos^{-1} \left(\frac{e^{2y} - 1}{e^{2y} + 1} \right) \equiv x \quad \frac{1}{2} \ln \frac{1 - \cos x}{1 + \cos x}$$

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$$\frac{y'}{y} = \frac{2x}{y^2}, y(2) = 3$$

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$$C_1 y = \tan^{-1}(x + 1) + \ln(3e^x + C) + C_1 y$$

$$+ \frac{1}{3} y = \ln(2e^x + 3) + \tan^{-1}(x - 3)$$

$$3 y = \cos^{-1}(x) \frac{e^{-y}}{2} = \frac{x^3}{2} - \frac{1}{2}, \sqrt[3]{x} - \frac{y}{2}$$

$$\frac{x^3}{3} y = \sqrt[3]{x}$$